

Improving Probabilistic Diffusion Models With Optimal Diagonal Covariance Matching

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May 4, 2026

Motivation

- Diffusion models generate by iteratively denoising Gaussian noise; mean is learned via score, variance often **fixed** or heuristically set.
- Learning **state-dependent diagonal covariance** improves sample quality and likelihood, particularly with few steps.
- Prior diagonal covariance learning is either via **VLB** or **data-driven** moments and can be biased or error-amplifying near $t \rightarrow 0$.
- This work: **OCM** directly regresses the **optimal diagonal analytic covariance** from the learned score with an **unbiased objective**.

Related Work

- Isotropic fixed covariances: **β -DDPM** and **$\tilde{\beta}$ -DDPM** [1]
- Isotropic state-independent: **A-DDPM** (analytic optimal within isotropic family) [3]
- State-dependent diagonal learned from data: **I-DDPM** (VLB) [2]
- Second-moment and residual learning: **SN-DDPM**, **NPR-DDPM** [4]
- Full optimal covariance identity from score: **Moment matching** [6]

Key gap

We want a **diagonal**, **state-dependent**, **score-based** covariance that is **unbiased**, accurate, and fast at inference.

Method: Optimal Covariance Identity

Generalized analytical covariance identity

Given $q(\tilde{x} | x) = \mathcal{N}(\alpha x, \sigma^2 I)$, the covariance of $q(x | \tilde{x})$ is

$$\Sigma(\tilde{x}) = \frac{\sigma^4 \nabla_{\tilde{x}}^2 \log q(\tilde{x}) + \sigma^2 I}{\alpha^2}.$$

- The optimal (full) covariance depends only on the **score Hessian**.
- In practice we use a **plug-in** learned score; full Hessian is prohibitive in high-D.

Goal

Learn the **diagonal Hessian** efficiently and unbiasedly, then map it to an optimal **diagonal covariance**.

Method: Diagonal Hessian and OCM Objective

Diagonal Hessian estimator

For $H(\tilde{x}) = \nabla_{\tilde{x}}^2 \log q(\tilde{x})$,

$$\text{diag}(H(\tilde{x})) \approx \frac{1}{M} \sum_{m=1}^M v_m \odot H(\tilde{x}) v_m, \quad v_m \sim \text{Rademacher}.$$

Unbiased Optimal Covariance Matching (OCM)

Train $h_\phi(\tilde{x})$ by

$$L_{\text{ocm}}(\phi) = \mathbb{E}_{q(\tilde{x})p(v)} \|h_\phi(\tilde{x}) - v \odot H(\tilde{x}) v\|_2^2,$$

which is an **upper bound** on the grounded objective and is **optimal** at $h_\phi = \text{diag}(H)$.

- In practice, $M = 1$ works well, making training efficient.

The bounded diagonal Hessian yields

Method: OCM in DDPM and DDIM

Skip-step DDPM (from t to t')

$$\mu_{t'}(x_t) = \frac{x_t + (1 - \bar{\alpha}_{t':t}) \nabla_{x_t} \log p_\theta(x_t)}{\sqrt{\bar{\alpha}_{t':t}}},$$

$$\Sigma_{t'}(x_t) = \frac{(1 - \bar{\alpha}_{t':t})^2 h_\phi(x_t) + (1 - \bar{\alpha}_{t':t}) I}{\bar{\alpha}_{t':t}}.$$

Skip-step DDIM (sample $x_0 \sim \mathcal{N}(\mu_0, \Sigma_0)$)

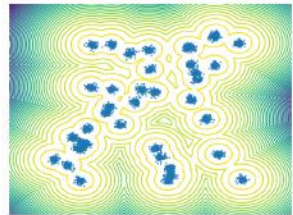
$$\mu_0(x_t) = \frac{x_t + (1 - \bar{\alpha}_t) \nabla_{x_t} \log p_\theta(x_t)}{\sqrt{\bar{\alpha}_t}}, \quad \Sigma_0(x_t) = \frac{(1 - \bar{\alpha}_t)^2 h_\phi(x_t) + (1 - \bar{\alpha}_t) I}{\bar{\alpha}_t}.$$

$$x_{t'} = \sqrt{\bar{\alpha}_{t'}} x_0 + \sqrt{1 - \bar{\alpha}_{t'}} \frac{x_t - \sqrt{\bar{\alpha}_t} x_0}{\sqrt{1 - \bar{\alpha}_t}}.$$

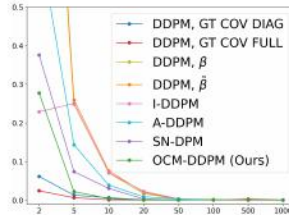
- OCM improves both **Markovian** (DDPM) and **non-Markovian** (DDIM) sampling.

Experimental Method

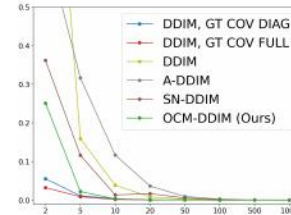
- Evaluate on **toy 2D MoG** and standard image datasets: CIFAR10 (LS, CS), CelebA 64x64, LSUN Bedroom, ImageNet 64x64, ImageNet 256x256 (latent/DiT).
- Metrics: **NLL** (ELBO upper bound), **FID**, **Recall** (for diversity).
- Use pre-trained score networks; train a small head h_ϕ with OCM; adopt covariance **clipping** at last steps per prior work.
- Compare against $\beta/\tilde{\beta}$ -DDPM, A-DDPM, I-DDPM, SN-DDPM, NPR-DDPM, DDIM variants.



(a) Toy MoG data



(b) DDPM MMD vs steps



(c) DDIM MMD vs steps

Experimental Setting

- Architectures: **UNet** for pixel space; **DiT** for latent ImageNet 256x256.
- Training: AdamW, learning rate 10^{-4} , 500K iters; select EMA checkpoint by validation FID.
- Sampling: Even trajectories, covariance clipping at final steps; report 50K-sample FID.



(a) CIFAR10

(b) CelebA 64

(c) LSUN Bed.

(d) ImageNet 256

Figure 2: Datasets used in experiments.

Experimental Results: Quality and Likelihood

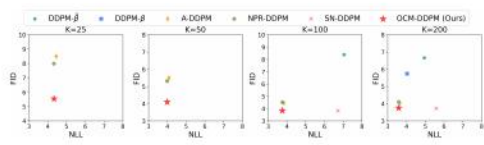


Figure 3: FID vs NLL trade-off on CIFAR10 (CS). OCM achieves the best balance.

Table 1: Least steps to reach FID ≈ 6 (lower is better).

Method	CIFAR10	CelebA	LSUN
DDPM	90	>200	130
DDIM	30	>100	>100
I-DDPM	45	—	90
A-DPM	25	55	100
NPR-DPM	23	50	90
SN-DPM	17	22	92
OCM-DPM	16	21	90

- OCM reduces steps needed for good FID and preserves strong likelihood.

Latent Diffusion on ImageNet 256x256 (DiT)

- Conditional generation with classifier-free guidance (CFG); 10-step sampling.
- Keep mean from pre-trained DiT, learn covariance via OCM.

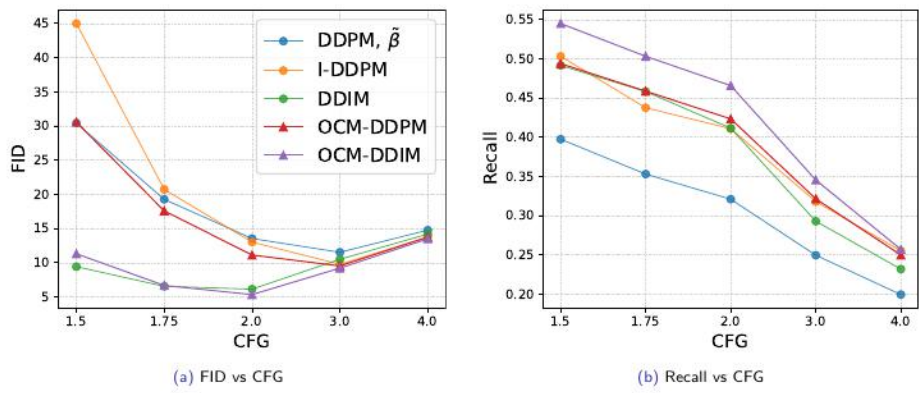


Figure 4: OCM achieves top FID at CFG=2.0 and better diversity (Recall).

Ablation: Rademacher Samples and Visualization

- Varying number of Rademacher samples M during OCM training shows minor gains; $M=1$ is sufficient in practice.

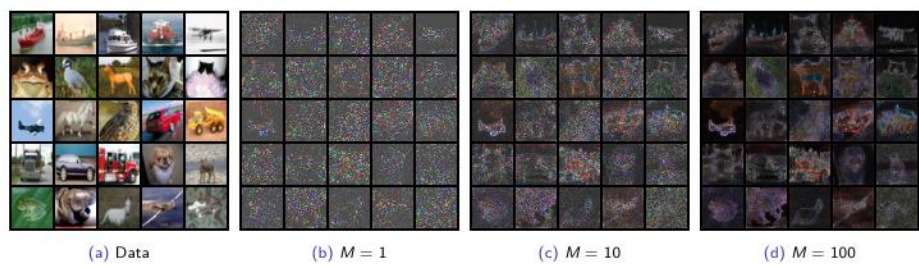


Figure 5: Diagonal covariance estimation visualization as M increases.

Deficiencies and Discussion

- Full covariance remains superior on some tasks, but is computationally infeasible in high dimensions.
- OCM learns only the diagonal; ignoring off-diagonals can leave performance on the table.
- Needs reliable scores; poor score estimates can degrade covariance quality.
- Clipping heuristics still matter at final steps.

Future Research

- Extend OCM to **low-rank/block-diagonal** structures for richer covariances at tractable cost.
- Combine with **ODE/SDE solvers** and **distillation** (e.g., moment-matched training).
- Apply to **video** and **inverse problems** for improved posterior sampling.
- Leverage OCM where **likelihood** is critical (e.g., compression, AI4Science sampling correction).

Thank You

- Paper: Improving Probabilistic Diffusion Models With Optimal Diagonal Covariance Matching
- Code: https://github.com/J-zin/OCM_DPM
- Contact: z.ou22@imperial.ac.uk, m.zhang@cs.ucl.ac.uk

Acknowledgments

ZO: Lee Family Scholarship. MZ and DB: AI Hub in Generative Models (EP/Y028805/1) and Cisco Centre of Excellence. Thanks to Wenlin Chen.

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